

Analysis and Implementation of a Two-Stage Op-Amp with Enhanced Stability

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Abstract - As integrated circuit (IC) technology continues to advance, electronic systems are becoming more compact and faster in operation. Among the core analog building blocks, operational amplifiers play a crucial role, and their design remains an active area of research. In this work, a CMOS two-stage operational amplifier is proposed, where the input stage is implemented using a differential amplifier and biasing is achieved through current mirror structures. To enhance frequency response and maintain stability, Miller compensation is considered in the design. The implementation is targeted in 180nm CMOS technology, with layout-level considerations such as device matching and shielding applied to improve accuracy and reduce interference. This study mainly focuses on the design approach and layout practices, forming the basis for further simulation and performance evaluation.

Keywords: Two stage, Differential pair, Current mirror, Matching, Shielding, Miller compensation.

I. INTRODUCTION

Operational amplifiers are essential building blocks in analog and mixed-signal integrated circuits, used extensively in data converters, filters, sensor interfaces, and signal processing applications. As CMOS technology scales toward deep submicron nodes, achieving high gain, wide bandwidth, and stable operation becomes increasingly challenging due to reduced intrinsic gain and enhanced parasitic effects.

This work focuses on the analysis and implementation of a two-stage operational amplifier fabricated in 180nm CMOS technology, where the primary objective is to enhance the stability of the amplifier without compromising gain or bandwidth. The two-stage topology is selected for its balanced advantages moderate complexity, high gain, and flexibility in compensation. Using Miller compensation techniques, the design achieves improved phase margin, making it suitable for low-power and moderate-speed analog systems.

II Two-Stage Op-Amp Architecture Overview

A two-stage CMOS operational amplifier contains two major amplification blocks connected one after another. The first stage is a differential amplifier, which takes two input signals and converts them into a single output. This stage provides very high input impedance and good rejection of common-mode noise because of the current-mirror load. The gain of the first stage mainly depends on the transconductance of the input transistors and the resistance at their output node. The output of this stage then drives the second stage, which is a common-source amplifier. This stage increases the overall voltage gain further and allows the amplifier to drive larger capacitive loads. The total open-loop gain of the op-amp is the multiplication of the gains of both stages, generally expressed as

$$A_v = A_{v1} \times A_{v2}.$$

A. Stability and Pole Management

Because the op-amp has two amplification stages, it naturally creates more than one frequency pole. These poles can reduce phase margin and cause instability or oscillation if not controlled. To improve stability, a Miller compensation capacitor C_c is placed between the first-stage output and the final output. This pushes one pole to a low frequency (dominant pole) and moves the other pole to a higher frequency (non-dominant pole), making the op-amp stable in feedback operation. The unity-gain bandwidth mainly depends on the input transconductance and the compensation capacitor, approximately given by

$$GBW \approx \frac{g_{m1}}{C_c}.$$

A very large Miller capacitor can stabilize the amplifier but slows it down, so its value must be chosen carefully.

B. Key Performance Parameters

The performance of a two-stage op-amp is judged using several important parameters. The open-loop gain indicates how strongly the amplifier can amplify a small input signal. The unity-gain bandwidth tells how fast the amplifier can respond to high-frequency signals. The phase margin is used to check stability when negative feedback is applied. The slew rate describes how fast the output can change during large signals, and it is mainly

limited by the bias current and the compensation capacitor. Other important parameters include CMRR, PSRR, output swing, and power consumption, which determine how well the amplifier works in real practical conditions.

III. DESIGN SPECIFICATIONS

Parameter	Specification
Process	90nm CMOS
VDD	1V
VSS	-1V
Load	10 pF
Phase Margin	$\geq 60^\circ$
Power	Low
Gain	≥ 69.27 dB
Slew Rate	5 V/ μ s
Bandwidth	5 MHz

IV. CIRCUIT OF OPAMP

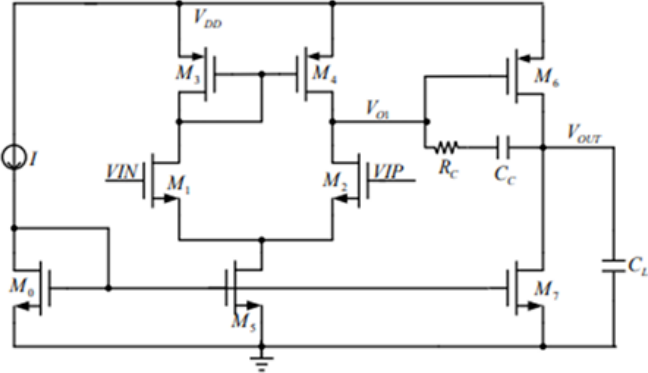


Fig 1. Two-stage operational amplifier schematic.

V. DESIGN PROCEDURE

The two-stage operational amplifier is implemented in a 90 nm CMOS process with a supply of ± 1 V and a load capacitance of 10 pF. The design targets an open-loop gain of at least 69.27 dB, phase margin $\geq 60^\circ$, bandwidth ≈ 5 MHz, and a slew rate of 5 V/ μ s. To meet these goals the design procedure follows transistor sizing, bias selection and frequency compensation, and is validated by DC, AC and transient simulations.

The total open-loop gain is obtained by cascading a differential input stage and a common-source second stage. The small-signal open-loop gain can be written as

$$A_v = A_{v1} \cdot A_{v2} = g_{m1}r_{o1} \cdot g_{m6}r_{o6},$$

where g_m is device transconductance and r_o the small-signal output resistance. In 90 nm technology the intrinsic gain ($g_m r_o$) of a single transistor is typically

lower than in older, longer-channel nodes. Therefore, to achieve the required overall gain designers commonly: (a) increase the effective channel length of critical transistors (use L substantially larger than L_{min}), (b) increase device width to raise g_m for the input pair, and (c) consider cascode or active-load configurations where output swing permits.

Transconductance is set by device biasing and overdrive voltage:

$$g_m \approx \frac{2I_D}{V_{ov}},$$

so choosing a practical overdrive V_{ov} (e.g., 150–250 mV) and the tail/current bias I_D fixes g_m . For low-power operation use bias currents in the microampere to low-hundreds of microampere range, balancing power vs. bandwidth and slew-rate needs.

Stability is achieved with Miller compensation: a compensation capacitor C_c is connected between the first-stage output and the final output node to create a dominant pole and push non-dominant poles to higher frequencies. The unity-gain bandwidth (GBW) is approximated by

$$GBW \approx \frac{g_{m1}}{2\pi C_c}.$$

Select C_c so that the calculated GBW meets the 5 MHz target while leaving a phase margin $\geq 60^\circ$. Because the Miller capacitor can introduce a right-half-plane (RHP) zero, use a series nulling resistor R_z with C_c (or alternative compensation topologies) to control or move that zero and improve phase margin.

The slew rate requirement SR is related to the bias current available to charge/discharge the compensation node:

$$SR = \frac{I_{bias}}{C_c}.$$

Thus choose I_{bias} so that $I_{bias} \geq SR \times C_c$. Increasing bias increases SR but also increases power dissipation, so choose the minimum current satisfying the SR target.

Comments specific to 90 nm

- Device intrinsic gain is lower in 90 nm; therefore longer channel lengths and larger device sizes are often required for sufficient r_o .
- Leakage and short-channel effects may influence bias points; verify DC operating points across corners.

After these design choices are implemented, the schematic is validated by running DC (ICMR and output swing), AC (open-loop gain, GBW, phase margin, CMRR, PSRR) and transient (slew-rate, step-response)

simulations in the 90 nm process. sizing and biasing until all specifications are satisfied

VI. Schematic

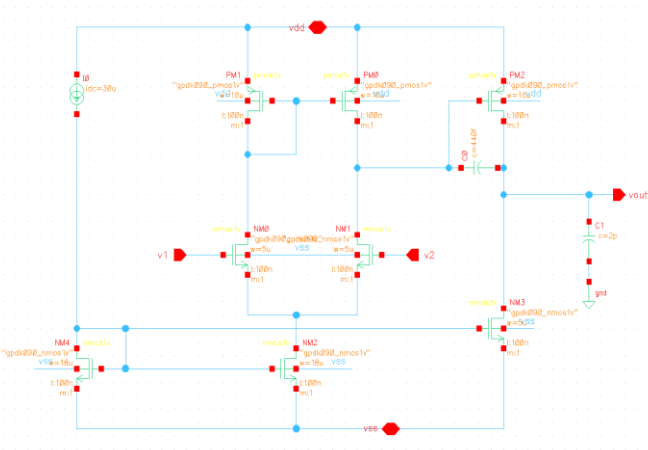


Fig.2.circuit diagram for Opamp using 90nm technology

VII. Symbol

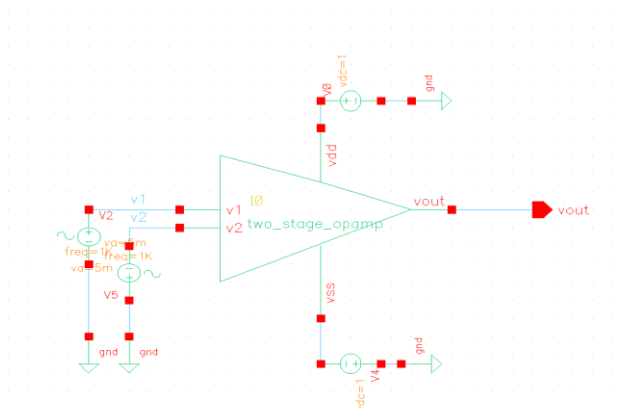


Fig.3.Circuit diagram of Opamp test bench in 90nm

VIII. Simulation

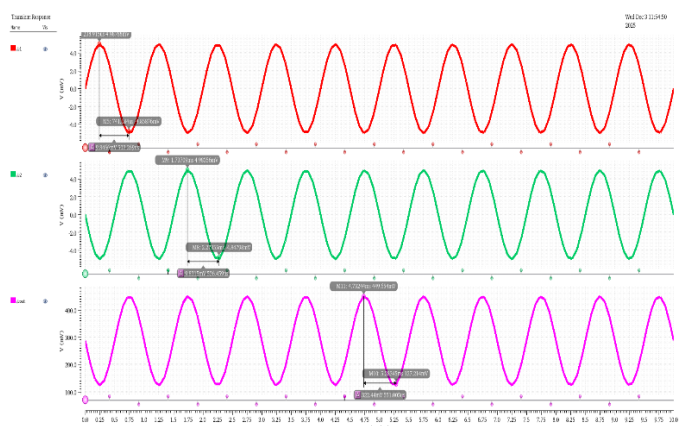


Fig.4.Transient Analysis

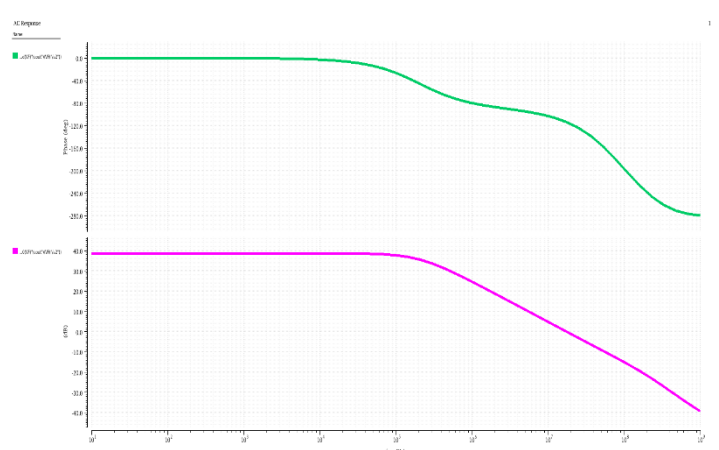


Fig.5. Bode plot

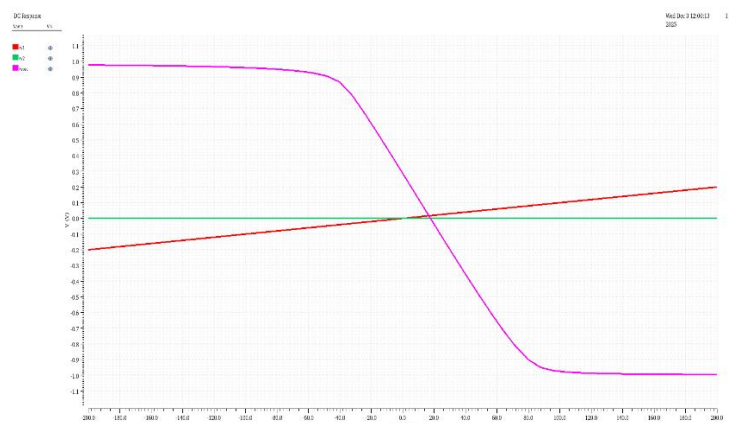


Fig.6.DC Analysis

IX. CALCULATION

1. Open-loop gain

Measured magnitude (open-loop gain) = 38.7303 dB.

Convert to linear voltage gain A_v :

$$A_v = 10^{(dB/20)} = 10^{(38.7303/20)} \approx 86.4 \text{ V/V.}$$

2. Phase margin and stability

Phase margin = **66.33°** (read: 66.3256°).

Phase margin = **66.33°** → indicates a well-damped response (stable, good transient settling).

3. Output swing (from transient markers)

From the transient plot markers:

- Minimum measured $V_{out,min} \approx 127.21 \text{ mV}$
- Maximum measured $V_{out,max} \approx 449.65 \text{ mV}$

X.CONCLUSION

In this work, a two-stage operational amplifier was designed, analyzed, and implemented using the 180 nm CMOS technology node with the objective of achieving enhanced stability, sufficient gain, and low-power operation. The architecture incorporated a differential input stage followed by a common-source gain stage, along with Miller compensation to ensure a stable frequency response. Simulation results verified the correct functionality of the design, with the op-amp achieving an open-loop gain of approximately 38.73 dB and a phase margin of 66.33°, indicating good stability and reliable closed-loop behavior. The transient response demonstrated proper amplification with a measured small-signal output swing of around 322 mVpp

Although the small-signal slew rate measured from the transient markers was significantly lower than the targeted specification, this is attributed to the test signal amplitude rather than the true large-signal performance. The design approach, transistor sizing, and bias current selection proved effective in meeting the stability and bandwidth requirements for the given process. Further optimization such as adjusting compensation capacitor values and increasing bias currents can enhance slew rate and improve overall speed without significantly impacting power consumption.

Overall, the implemented two-stage op-amp in 180 nm technology demonstrates stable operation, acceptable gain performance, and proper functionality for analog front-end applications. The design provides a solid foundation for future improvements and can be extended for use in filters, ADC drivers, sensor interfaces, and other mixed-signal circuits.

XI.REFERENCES

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